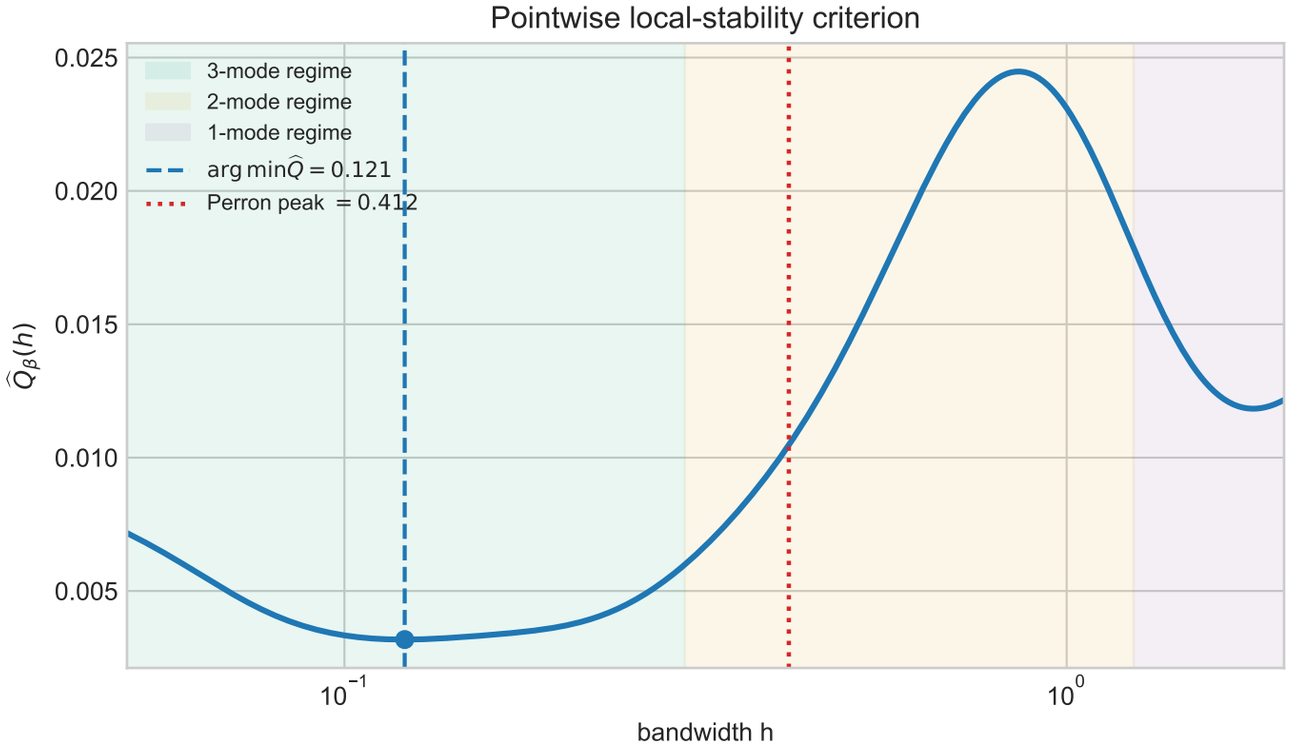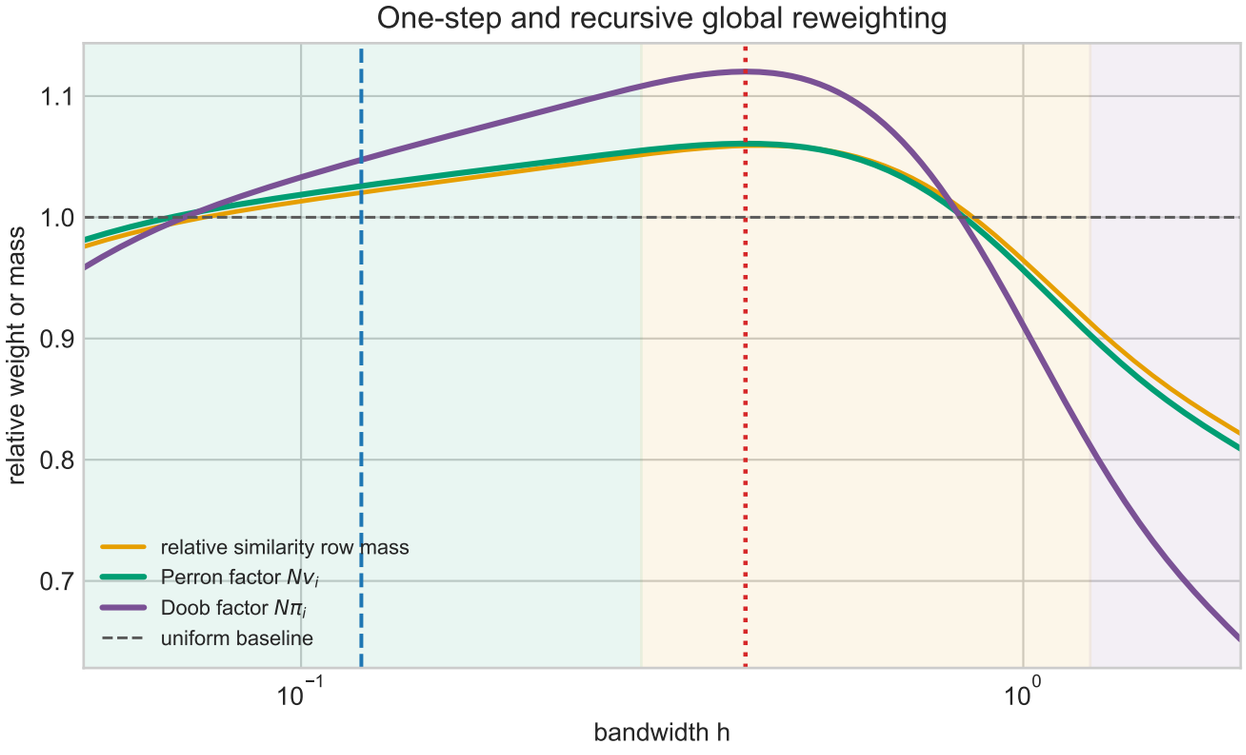Local stability versus global similarity reweighting